

**Hivelet: A Web-Based Apartment Management System for
Fe Galang Da Silva Boarding House**

An Undergraduate Capstone Project
presented to the
Information Technology Department
Bicol University College of Science
Legazpi City

In Partial Fulfillment of the
Requirements for the Degree of
Bachelor of Science in Information Technology

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RECOMMENDATION FOR THE ORAL DEFENSE

The undergraduate research project entitled, **“Hivelet: A Web-Based Apartment Management System for Fe Galang Da Silva Boarding House”**, prepared and submitted by **John Lloyd M. Cuario, Eljohn Paulo C. Loterte, Victor Noel A. Napay, Sean Jerve LI. Rebancos, Kiel Hedrix V. Relos**, in partial fulfillment of the requirements for the degree **BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY**, is hereby submitted to the research project committee for oral examination.

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In partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology, this research project entitled, **“Hivelet: A Web-Based Apartment Management System for Fe Galang Da Silva Boarding House”**, prepared and submitted by **John Lloyd M. Cuario, Eljohn Paulo C. Loterte, Victor Noel A. Napay, Sean Jerve LI. Rebancos, Kiel Hedrix V. Relos**, is hereby recommended for Oral Examination.

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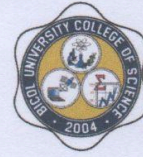
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CONTENT ADVISER'S CERTIFICATION

This is to certify that this undergraduate research project entitled, **"Hivelet: A Web-Based Apartment Management System with Financial Analytics, Booking, and Issue Tracking for Fe Galang Da Silva Boarding House"**, prepared and submitted by **Cuario, John Lloyd M, Loterte, Eljohn Paulo C., Napay, Victor Noel A., Rebancos, Sean Jerve LI., Relos, Kiel Hedrix V.**, in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology, has been read and edited by the undersigned.

Issued this 30th day of April, 2026 at Bicol University College of Science, Legazpi City.


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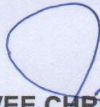
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PROGRAMMING ADVISER'S CERTIFICATION

This is to certify that this undergraduate research project entitled, **"Hivelet: A Web-Based Apartment Management System with Financial Analytics, Booking, and Issue Tracking for Fe Galang Da Silva Boarding House"**, prepared and submitted by **Cuario, John Lloyd M, Loterte, Eljohn Paulo C., Napay, Victor Noel A., Rebancos, Sean Jerve Ll., Relos, Kiel Hedrix V.**, in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology, has been evaluated by the undersigned.

Issued this **30th** day of **April, 2026** at Bicol University College of Science, Legazpi City.


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ABSTRACT

TABLE OF CONTENTS

	page
RECOMMENDATION FOR ORAL DEFENSE	ii
RESULT OF FINAL ORAL DEFENSE	iii
APPROVAL SHEET	iv
CONTENT ADVISER'S CERTIFICATION	v
PROGRAMMING ADVISER'S CERTIFICATION	vi
ACKNOWLEDGEMENT	vii
ABSTRACT	vii
TABLE OF CONTENTS	ix
LIST OF TABLES	xii
LIST OF FIGURES	xiii
1 INTRODUCTION	1
1.1 Background of the Study	1
1.2 Objectives of the Study	4
1.3 Significance of the Study	5
1.4 Scope and Delimitations	6
2 THEORETICAL FRAMEWORK	10
2.1 Review of Related Literature	10
2.1.1 Structural Inefficiencies in Property Management Systems	10
2.1.2 Management Information Systems as the Structural Foundation	11
2.1.3 Financial Analytics and Operational Decision Support	12

2.1.4 Digital Communication and Issue Resolution Systems	13
2.1.5 Progressive Web Application as an Integration Enabler	13
2.1.6 Synthesis of Literature	14
2.2 Gap Bridged by the Study	15
2.3 Concept of the Study	16
2.4 Definition of Terms	17
3 MATERIALS AND METHODS	21
3.1 Materials	21
3.1.1 Software	21
3.1.2 Hardware	23
3.2 Software Methodology	24
3.2.1 Requirements	25
3.2.2 Design	26
3.2.3 Development	26
3.2.4 Testing	27
3.2.5 Deployment	27
3.2.6 Review and Maintenance	28
3.3 Evaluation Procedure	28
4 RESULTS AND DISCUSSION	30
4.1 Objective 1 Title	30
4.1.1 Sub-objective Title	30

4.2 Objective 1 Title	31
4.2.1 Sub-objective Title	31
4.3 Evaluation of the Developed Pet Adoption System	31
Evaluation Results	31
Deployment Plan and Strategies	32
5 SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS	33
5.1 Summary	33
5.2 Conclusions	35
5.3 Recommendations	35
REFERENCES	35
APPENDICES	65
Appendix A: Letter of Request	65
Appendix B: Appointments	66
Appendix C: Gantt Chart	70
Appendix D: Data Flow Diagram (Context Flow Diagram)	71
Appendix E: Data Flow Diagram (Diagram 0)	72
Appendix F: <u>Data Flow Diagram</u> (Child Diagrams)	73
Appendix G: Use Case Diagram of the Proposed System	75
Appendix H: Entity Relationship Diagram of the Proposed System	76
Appendix I: Initial Design of the Proposed System	86
Appendix J: Evaluation Guide	10

Appendix K: User's Manual	107
Appendix L: How To Test (Test Cases) <System Name>	108
Appendix M: Source Code	110
CURRICULUM VITAE_x	150

LIST OF TABLES

Table 1. Software Requirements	28
Table 2. Hardware Requirements	29
Table 3. Mobile Testing Hardware Specifications (Baseline)]	30
Table 4. Likert Scale	36
Table 5. Summary of Evaluation Results	45
Table 6. Evaluation Results Based from Functional Suitability	45
Table 7. Evaluation Results Based from Usability	46
Table 8. Evaluation Results Based from Reliability	47
Table 9. Software and Hardware Requirements for Deployment	50
Table 10. Deployment Plan and Strategies	51

LIST OF FIGURES

Figure 1. Management Information System (MIS) framework by Laudon and Laudon	13
Figure 2. Conceptual Framework	22
Figure 3. Agile Software Development Life Cycle	31

1 INTRODUCTION

1.1 Background of the Study

Administrative processes in apartment management typically involve tenant record keeping, financial tracking, and maintenance coordination, which become increasingly complex as the number of units increases (Burke, 2026). While web-based property management systems have been developed to support these functions, their adoption in small-scale residential settings remains limited due to operational constraints such as limited infrastructure, technical capacity, and resource availability (Mia et al., 2024).

In small-scale boarding house operations, administrative tasks are still managed using standalone spreadsheet files and manual documentation. In observed cases within the study context, these records are stored in removable storage devices, exposing the system to risks in data integrity such as accidental deletion, file corruption, duplication, and lack of version control. In addition, tenant concerns and maintenance requests are often communicated through informal channels such as messaging applications or verbal reporting, resulting in untracked requests, delayed responses, and inconsistent follow-up.

These conditions reflect fragmented administrative workflows, where records, communication, and transactions are managed across separate, non-integrated tools. Consequently, property owners may experience difficulties in maintaining accurate records, monitoring financial transactions, and ensuring

timely resolution of tenant concerns. Although existing web-based property management systems aim to address such inefficiencies through centralized data storage and real-time processing (Kotkar et al., 2026), many are commonly designed with assumptions aligned to large-scale or commercial property environments. These systems typically assume stable infrastructure, structured workflows, and dedicated administrative personnel (Ullah et al., 2021), which do not reflect the operational conditions of small-scale, resource-constrained boarding houses.

In the Philippine context, these challenges are further reinforced by reliance on cash-based transactions, limited access to integrated digital tools, and continued dependence on manual record-keeping practices (Mia et al., 2024). As a result, many small-scale apartment operations continue to function under fragmented and informal administrative processes despite the availability of existing digital solutions.

These limitations are also reflected in local implementations. Magno et al. (2024) developed an apartment rental management system to address issues such as disorganized data management, difficulty in tracking tenant payments, missed billing deadlines, and reliance on paper-based records. Their findings demonstrate that digitizing transaction tracking and task organization improves administrative efficiency and data accuracy. However, the system primarily focuses on financial transactions and task management, without integrating communication, booking, and tenant interaction workflows into a unified system.

Similarly, Pavia et al. (2022) developed a property management system for Hospitality Management students at Bicol College, demonstrating how digital platforms can enhance coordination, record-keeping, and service management within a controlled academic environment with predefined workflows and institutional support. While the study confirms the effectiveness of system-based management, these conditions differ significantly from small-scale boarding house operations, where administrative processes are informal, decentralized, and constrained by limited resources.

Taken together, these local studies demonstrate that while digital solutions can improve specific aspects of property management, they are often implemented either as function-specific systems or within controlled environments. As such, they do not fully address the fragmented and non-integrated nature of administrative workflows in real-world, resource-constrained boarding house settings.

Many existing systems therefore remain misaligned with the operational realities of small-scale, apartment management environments. This reveals a critical gap: the absence of an integrated platform that consolidates tenant management, financial tracking, communication, and booking within a single, context-appropriate system.

This study proposes Hivelet, a web-based apartment management system designed to address these limitations by centralizing administrative processes and providing a structured mechanism for managing tenant records, financial

transactions, and maintenance concerns within a unified platform, specifically tailored for resource-constrained residential settings.

1.2 Objectives of the Study

The primary objective of this study is to address the mismatch between existing property management systems and the operational realities of small-scale, resource-constrained boarding houses by designing, developing, and evaluating *Hivelet*, a web-based apartment management system that improves administrative workflow integration, communication, and operational control. Specifically, the study aims:

1. To analyze the existing apartment management practices of the Fe Galang Da Silva Apartment to determine the challenges in tenant management, financial tracking, communication, and booking processes, and establish system requirements based on the identified operational gaps and user needs.
2. To develop the Hivelet system to feature tenant and room management, booking and reservation management, financial tracking with optional online payment recording, a maintenance ticketing and notification system, role-based access control for administrators and tenants, and Progressive Web Application (PWA) support for responsive cross-device accessibility.

3. To pilot test the developed Hivelet system for functionality, responsiveness, and operational performance.
4. To evaluate and optimize the system based on the ISO/IEC 25010 software quality model in terms of: Functional suitability, Performance efficiency, Compatibility, Usability, Reliability, Security, Maintainability, Portability

1.3 Significance of the Study

The development of Hivelet, a web-based apartment management system implemented as a Progressive Web Application (PWA), is significant in addressing the operational challenges of fragmented and manual administrative workflows in small-scale, resource-constrained boarding house environments.

For **property administrators and owners**, the study provides a centralized system that reduces dependence on manual record-keeping practices such as spreadsheets and removable storage devices, which are prone to data inconsistency, loss, and duplication. By integrating tenant records, financial transactions, and maintenance concerns into a unified platform, the system supports more reliable information management and improves administrative coordination in environments affected by cash-based transactions and limited digital infrastructure.

For **tenants**, the study improves communication efficiency by providing a structured mechanism for submitting and tracking maintenance concerns and viewing relevant account information. This reduces reliance on informal communication channels such as messaging applications and verbal reporting, which often result in delayed responses and untracked requests.

For **future researchers**, the study contributes to a localized implementation of a Progressive Web Application–based property management system designed specifically for small-scale residential contexts. The system architecture, integration approach, and evaluation using the ISO/IEC 25010 model may serve as a reference for further research, particularly in developing systems that address the gap between enterprise-level property management solutions and resource-constrained residential operations.

1.4 Scope and Delimitations

This study focuses on the design, development, and evaluation of Hivelet, a Progressive Web Application (PWA) created to transition the 32-unit Fe Galang Da Silva Apartment near Bicol University from manual, paper-based administration to a centralized digital management system. To achieve this, Hivelet integrates comprehensive room and tenant management capabilities, facilitating seamless onboarding, the secure digital storage of records, real-time monitoring of bed availability, and a public-facing booking module that allows

prospective renters to submit reservation inquiries without requiring full system access. Furthermore, the application features a financial tracking component dedicated to monitoring cash flow, expenses, and tenant payments; while this primarily supports manual, cash-based transaction recording, it offers an optional digital payment submission feature for added convenience. Operational communication is also streamlined through a structured issue ticketing system, enabling tenants to log maintenance requests that administrators can systematically assign, track, and update until resolution.

Despite these comprehensive features, the scope is bound by specific delimitations regarding system accessibility and financial processing: although the PWA utilizes cached resources for limited offline access and responsive design for diverse devices, core functionalities—such as data encoding, real-time synchronization, and booking updates—strictly require an active internet connection to maintain data consistency. Additionally, the supplementary online payment feature is restricted to basic transaction recording and explicitly excludes advanced financial processing capabilities like automated reconciliation, refunds, or direct banking integration. Ultimately, to ensure the system effectively resolves the apartment's operational inefficiencies, Hivelet will be rigorously evaluated against the ISO/IEC 25010 software quality standard, with a specific emphasis on measuring its functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

2 THEORETICAL FRAMEWORK

This chapter establishes the theoretical and empirical foundation of the Hivelet system by synthesizing literature on property management systems, Management Information Systems (MIS), financial analytics, digital communication systems, and Progressive Web Applications (PWA). Rather than treating these domains independently, the study positions them as interconnected operational components within apartment management environments. The framework argues that inefficiencies in small-scale residential operations emerge primarily from fragmented administrative structures, where records, financial monitoring, communication, and operational coordination exist across disconnected tools and processes. The reviewed literature collectively supports the need for an integrated, centralized, and accessible management system tailored for resource-constrained apartment environments.

2.1 Review of Related Literature

2.1.1 Fragmentation and Operational Inefficiencies in Property Management

The increasing complexity of apartment and boarding house operations has intensified the need for structured and centralized management systems. Existing literature consistently identifies fragmentation of administrative workflows as one of the primary operational issues in residential property management environments (Ullah et al., 2021; Burke, 2026; Setty, 2022). As the number of tenants, transactions, and maintenance activities increases, traditional methods such as manual documentation, spreadsheet-based tracking, and

informal communication becomes increasingly difficult to manage consistently and accurately.

Ullah et al. (2021) explain that conventional property management practices often suffer from disconnected workflows, documentation inconsistencies, and inefficient coordination between operational tasks. Similarly, Burke (2026) emphasizes that increasing residential density proportionally increases administrative complexity, particularly in areas involving tenant monitoring, maintenance coordination, and financial recording. These findings collectively suggest that operational inefficiencies are not isolated technical issues but structural problems caused by fragmented information handling and disconnected administrative processes.

This condition is further reflected in localized implementations. Magno et al. (2024) observed that apartment administrators commonly experience difficulties in tracking payments, organizing tenant records, and managing paper-based transactions. Their findings demonstrate that digitization improves data consistency and transaction monitoring; however, the study also reveals that partial digital implementation alone does not fully eliminate fragmentation when operational functions remain separated. Likewise, Pavia et al. (2022) demonstrate that system-based management improves organizational coordination and service operations, yet their implementation operates within a controlled institutional environment with predefined workflows and technical support, conditions that differ significantly from resource-constrained apartment operations.

Collectively, these studies establish that the primary challenge in apartment management is not merely the absence of digital systems, but the absence of integrated operational structures capable of consolidating administrative functions into a unified environment. This reinforces the need for centralized information management approaches capable of organizing fragmented workflows into cohesive and traceable system processes.

2.1.2 Management Information Systems as the Foundation of Centralization

This theoretical foundation is grounded in Management Information Systems (MIS), which emphasize the transformation of fragmented operational data into centralized and actionable information for decision-making. According to Laudon and Laudon, MIS functions as an organized system that collects, processes, stores, and distributes information to support operational control, coordination, and administrative decision-making. Within apartment management environments, this framework explains why manual and semi-digital operations often become inefficient, as information is typically distributed across isolated records, disconnected tools, and inconsistent communication channels.

This relationship is further illustrated in Figure 1, based on the Management Information System (MIS) framework by Laudon and Laudon, which demonstrates the flow of data from fragmented operational sources through system processing mechanisms, resulting in consolidated and meaningful information outputs that support organizational decision-making.

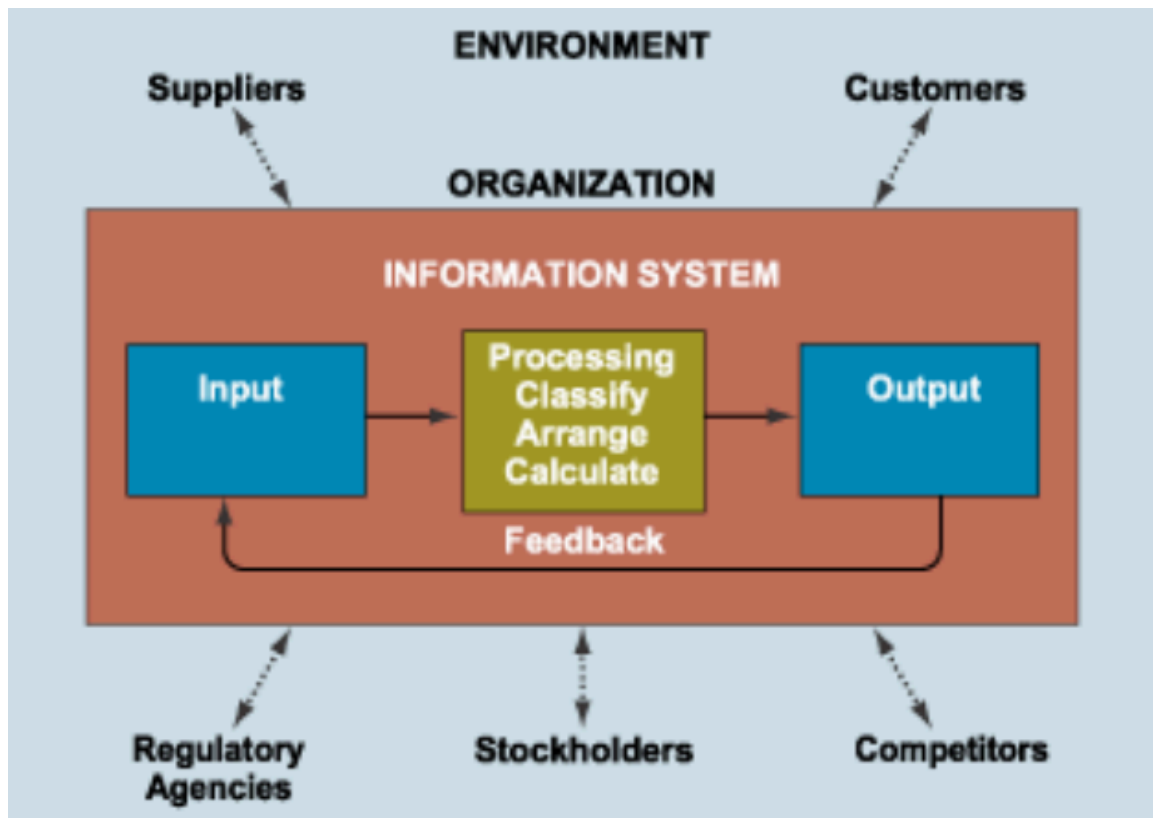


Figure 1. Management Information System (MIS) framework by Laudon and Laudon

The importance of centralized information processing is further reinforced by Ghasemaghaei et al. (2018), who emphasize that organizational decision-making performance significantly improves when operational data is structured, integrated, and analytically accessible. Their findings indicate that data analytics competency enables more accurate forecasting, operational monitoring, and financial oversight, particularly in environments requiring continuous administrative coordination. Similarly, Pressman and Maxim (2020) explain that integrated software systems improve consistency, maintainability, and operational reliability by reducing redundancy and streamlining data flow across system components.

Within apartment management operations, these principles directly support the need for centralized architectures capable of integrating tenant information, maintenance records, communication logs, and financial transactions within a single operational environment. The literature therefore positions MIS not simply as a technological concept, but as the structural mechanism through which fragmented administrative activities become coordinated, traceable, and manageable.

This theoretical perspective also aligns with findings from property management studies which indicate that operational inefficiencies often emerge when information systems lack interoperability and centralized accessibility (Burke, 2026; Magno et al., 2024). Consequently, the integration of operational data within a unified system environment becomes essential for improving administrative visibility, reducing redundancy, and supporting informed decision-making processes.

2.1.3 Financial Analytics and Administrative Decision Support

Financial management serves as one of the most critical operational components within apartment administration, as payment monitoring, expense tracking, and cash flow visibility directly influence managerial decision-making. Existing studies consistently emphasize that financial inefficiencies often emerge when payment records are maintained through fragmented or manual processes (Encarnacion et al., 2025; Magno et al., 2024; Setty, 2022).

Encarnacion et al. (2025) demonstrate that structured digital payment monitoring systems significantly improve administrative oversight within student boarding house operations by providing organized financial ledgers, payment summaries, and transaction visibility. Their findings suggest that centralized financial records reduce monitoring inconsistencies and improve forecasting accuracy. Similarly, Magno et al. (2024) observed that digital transaction tracking improves billing management and minimizes missed payment monitoring compared to paper-based systems. Setty (2022) further explains that centralized financial management systems improve transparency and reduce administrative errors by consolidating transaction records into a unified database environment.

These findings strongly align with MIS principles regarding centralized information processing and structured data accessibility. Ghasemaghaei et al. (2018) argue that analytical decision-making depends heavily on the quality, consistency, and accessibility of operational data. In apartment management contexts, financial analytics therefore functions not merely as a reporting mechanism, but as a decision-support component that enables administrators to monitor operational performance, identify irregularities, and maintain financial accountability.

Collectively, the literature suggests that financial monitoring becomes significantly more effective when integrated into a centralized operational framework rather than implemented as a standalone feature. This reinforces the necessity of integrating financial analytics within broader apartment management

systems to support consistent administrative control and operational transparency.

2.1.4 Digital Communication and Structured Issue Resolution

Beyond financial management, operational efficiency in apartment environments is also heavily influenced by communication quality and maintenance coordination processes. Existing literature identifies informal communication practices as major contributors to delayed responses, inconsistent issue tracking, and reduced tenant satisfaction (Jing & Lim, 2021; Saputra et al., 2025).

Jing and Lim (2021) explain that management service quality significantly affects residential satisfaction and operational effectiveness within apartment environments. Their study highlights that responsiveness, communication consistency, and issue resolution efficiency are critical components of effective property administration. However, in traditional management settings, tenant concerns are often communicated through verbal reporting or messaging applications, resulting in undocumented requests and inconsistent follow-up procedures.

To address these issues, Saputra et al. (2025) demonstrate that web-based room management systems with structured issue coordination mechanisms significantly improve operational accountability and response management. By transforming tenant complaints into recorded and traceable

maintenance workflows, digital ticketing systems improve monitoring consistency and reduce communication gaps between administrators and tenants.

These findings collectively indicate that communication systems should not be treated merely as supplementary features but as integral operational components within apartment management environments. Consistent with MIS theory, structured communication mechanisms improve information traceability, coordination efficiency, and administrative visibility by ensuring that operational concerns are documented within centralized system workflows rather than dispersed across informal channels.

The literature therefore supports the integration of maintenance ticketing and structured communication modules within unified apartment management systems to improve accountability, responsiveness, and operational coordination.

2.1.5 Progressive Web Applications as an Accessibility and Integration Framework

While centralized systems improve operational coordination, effective implementation also depends on accessibility across devices and varying infrastructure conditions. Progressive Web Applications (PWAs) provide a technological framework capable of addressing these accessibility challenges within resource-constrained environments (Google, 2018; Biørn-Hansen et al., 2019).

Google (2018) describes PWAs as web applications that combine the accessibility of browser-based systems with functionalities commonly associated with native mobile applications, including responsiveness, offline capability, and cross-device compatibility. Similarly, Biørn-Hansen et al. (2019) explain that PWAs reduce development complexity by enabling a single application architecture to operate consistently across multiple platforms without requiring separate native implementations.

These characteristics are particularly relevant within small-scale apartment operations where infrastructure limitations, inconsistent connectivity, and device diversity may restrict adoption of conventional enterprise systems. Mia et al. (2024) further emphasize that small and medium-sized enterprises in the Philippines commonly experience barriers related to infrastructure limitations, technical capacity, and resource availability during digital technology adoption. This suggests that accessibility and implementation simplicity are critical considerations in system deployment within localized operational environments.

From a systems perspective, PWA architecture functions as the technological delivery layer that enables centralized MIS functions to remain accessible and operational across varying conditions. Rather than serving solely as a development approach, PWAs support the broader objective of operational integration by ensuring that administrative tools, communication workflows, and financial monitoring systems remain consistently accessible across devices and environments.

2.1.6 Synthesis of Literature

The reviewed literature collectively demonstrates that operational inefficiencies in apartment management environments primarily emerge from fragmented administrative structures rather than the complete absence of digital technologies. Property management studies identify persistent issues involving disconnected workflows, inconsistent record management, delayed communication, and operational redundancy (Burke, 2026; Ullah et al., 2021; Magno et al., 2024). MIS literature explains these inefficiencies as consequences of fragmented information processing and lack of centralized operational visibility (Ghasemaghaei et al., 2018; Pressman & Maxim, 2020).

Financial analytics studies further establish that structured and centralized financial data significantly improves administrative oversight, forecasting capability, and operational transparency (Encarnacion et al., 2025; Setty, 2022). Similarly, communication and maintenance coordination studies emphasize that operational accountability improves when issue resolution processes are formalized through structured digital workflows rather than informal communication channels (Jing & Lim, 2021; Saputra et al., 2025).

Meanwhile, PWA literature provides the technological framework that enables these integrated operational components to remain accessible within environments characterized by limited infrastructure and varying device conditions (Google, 2018; Biørn-Hansen et al., 2019; Mia et al., 2024).

Despite their complementarity, these studies are commonly implemented in isolation. Existing systems frequently focus on specific operational functions such as financial management, communication, or tenant tracking independently, resulting in partial solutions that fail to address the interconnected nature of apartment management workflows. Consequently, the literature reveals a need for an integrated operational framework that centralizes administrative processes, formalizes communication, supports financial monitoring, and remains accessible within resource-constrained residential environments.

2.2 Gap Bridged by the Study

Based on the synthesis of literature, a significant research gap emerges regarding the absence of an integrated apartment management system specifically designed for small-scale and resource-constrained residential operations. Existing systems are commonly developed either as enterprise-oriented platforms or as isolated functional solutions focusing on individual administrative tasks such as payment monitoring, tenant management, or maintenance coordination (Magno et al., 2024; Setty, 2022; Saputra et al., 2025).

While these systems demonstrate improvements in specific operational areas, they frequently assume stable infrastructure, dedicated administrative personnel, and structured operational environments that do not reflect the realities of small-scale apartment and boarding house operations (Mia et al.,

2024; Burke, 2026). As a result, many administrators continue to rely on fragmented workflows involving manual documentation, spreadsheet-based records, informal communication channels, and disconnected transaction monitoring processes.

Furthermore, although studies support the importance of centralized information systems, financial analytics, structured communication, and accessible deployment architectures, few implementations integrate these components into a single operational framework tailored for localized, resource-constrained residential settings.

This study addresses this gap through the development of Hivelet, a Progressive Web Application-based apartment management system designed specifically for small-scale apartment operations. The system integrates centralized MIS principles, financial tracking mechanisms, structured issue ticketing, tenant management, booking coordination, and accessible cross-device deployment within a unified operational environment.

By consolidating these interconnected administrative functions into a single platform, the study transforms fragmented workflows into cohesive, traceable, and structured system processes, thereby aligning theoretical principles with practical implementation within real-world residential management contexts.

2.3 Concept of the Study

This study is grounded on the concept of improving apartment management by integrating its core processes into a centralized digital platform. Apartment operations involve interconnected activities such as tenant management, payment tracking, booking coordination, and maintenance handling. In traditional setups, these processes are often managed manually and separately, resulting in delays, inconsistencies, and increased administrative workload.

Figure 1 illustrates the study's conceptual framework using the Input-Process-Output (IPO) model, a standard software engineering approach that maps how system components interact to transform inputs into outputs.

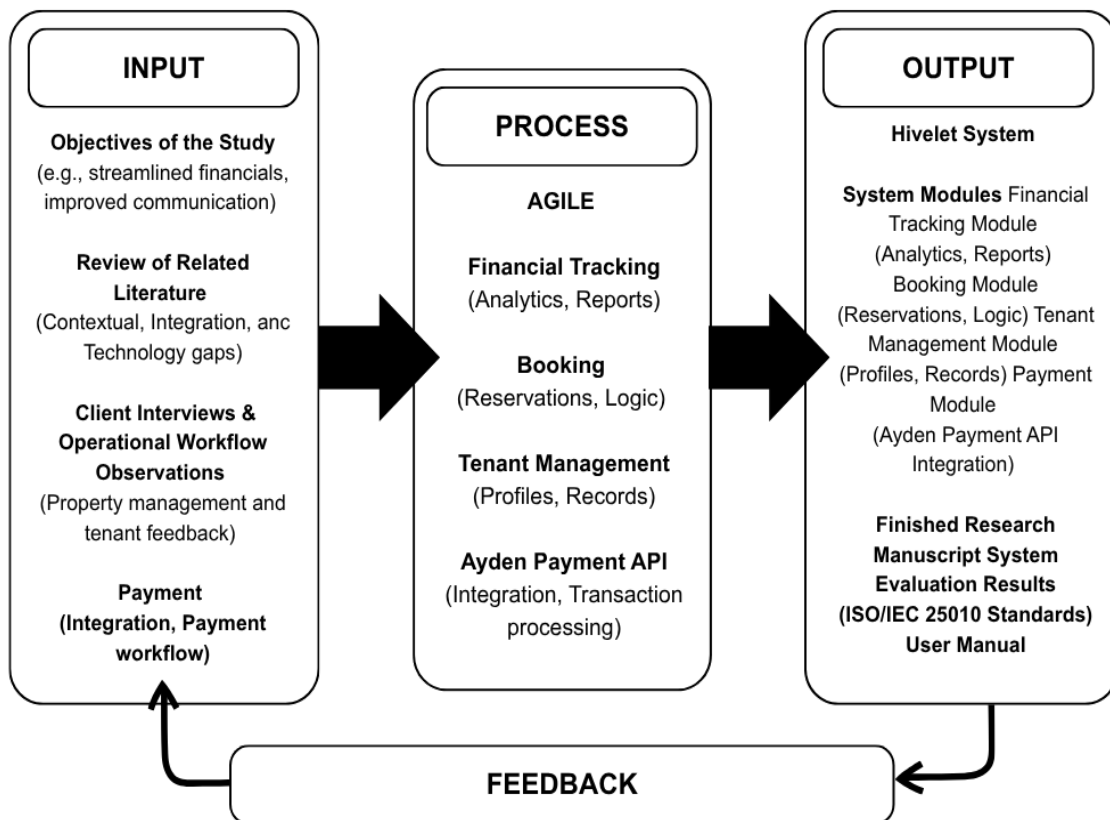


Figure 2. Conceptual Framework

In this study, the Input phase consists of the study objectives, review of related literature, client interviews, needfinding observations, and analysis of existing administrative and payment workflows. These inputs provide both theoretical and contextual foundations, allowing the researchers to identify operational gaps, user requirements, and system constraints within the apartment management environment. The Process phase applies the Agile Software Development Life Cycle, integrating iterative stages of planning, design, development, testing, deployment, and review. Within this phase, the system modules are implemented, including financial tracking (analytics and reports), booking management (reservation logic), tenant management (profiles and records), and an optional online payment integration using the Adyen API. These processes transform identified requirements into a functional and structured system aligned with real-world operational needs. The Output phase includes the developed Hivelet system, its functional modules, and generated outputs such as financial reports, booking records, and transaction receipts. It also includes the completed research manuscript, system evaluation results based on ISO/IEC 25010 standards, and the user manual. These outputs reflect the system's effectiveness in improving administrative efficiency, data organization, and communication within the apartment management process. Feedback is incorporated across all phases, allowing continuous system refinement based on user evaluation and operational performance, ensuring that the system remains adaptive and responsive to real-world conditions.

2.4 Definition of Terms

Hivelet – A web-based apartment management system developed as a Progressive Web Application (PWA) for the Fe Galang Da Silva Apartment, designed to centralize and integrate tenant management, financial tracking, booking inquiries, and maintenance issue reporting into a unified operational platform that supports small-scale, resource-constrained administrative environments.

Progressive Web Application (PWA) – A web-based application architecture that enables app-like functionality through browsers, providing responsive design, offline capability, and cross-device accessibility without requiring installation, making it suitable for low-resource and hybrid operational environments.

Resource-Constrained Environment – Refers to small-scale residential or boarding house operations characterized by limited financial resources, minimal technical infrastructure, and constrained administrative manpower, where management processes are typically dependent on hybrid manual and basic digital tools rather than fully integrated systems.

Fragmented Workflow – A condition in which administrative processes such as tenant management, financial recording, and maintenance reporting are handled using disconnected tools (e.g., spreadsheets, paper records, and messaging applications), resulting in inconsistent data, delayed coordination, and reduced operational efficiency.

Centralized System – A system architecture in which all operational data and processes are managed within a single integrated platform, enabling consistent data storage, real-time access, and unified control over administrative functions.

Small-Scale Apartment Management – Refers to the administrative operation of residential boarding houses or apartments with limited units, minimal staff, and constrained digital infrastructure, where management processes are often hybrid, combining manual and basic digital tools.

Management Information System (MIS) – An information system framework that processes raw operational data into structured and meaningful information to support decision-making, coordination, and control within an organization or operational environment.

Booking Inquiry – A system feature that allows prospective tenants to submit room availability requests through the platform, which are reviewed and processed by administrators for approval or scheduling purposes.

Financial Tracking – A system function that records and monitors tenant payments and operational expenses in a structured digital format, ensuring accurate financial records and supporting administrative monitoring of income and expenditures.

Cash Flow Monitoring – The process of generating summarized financial reports based on recorded transactions, providing insights into income, expenses, and overall financial movement within the apartment system.

Issue Ticketing System – A structured digital mechanism that allows tenants to submit maintenance or facility-related concerns, which are then tracked, assigned, and resolved by administrators through a centralized workflow system.

User Authentication – A security process that verifies the identity of users accessing the system to ensure authorized entry and protect sensitive operational data.

Role-Based Access Control (RBAC) – A security model that restricts system access based on user roles (e.g., admin, tenant), ensuring that users can only access functions relevant to their responsibilities.

Optional Online Payment Feature – A supplementary system function that allows tenants to input or simulate digital payment transactions within the platform, without serving as the primary financial processing mechanism of the system.

ISO/IEC 25010 – An international software quality standard used to evaluate system quality based on attributes such as functional suitability, usability, and security, ensuring structured assessment of system performance.

3 MATERIALS AND METHODS

This chapter presents the strategies, tools, and methods to be utilized in the development of the Hivelet system. It also outlines the instruments and procedures that will be used in evaluating the system's functionality, performance, and user acceptability.

3.1 Materials

This section presents the details of the materials to be utilized in the documentation of the study and in the development of the Hivelet system. These materials include both software and hardware components essential for the design, development, testing, and deployment of the proposed Progressive Web Application (PWA). The software tools include programming languages, frameworks, and database technologies, while the hardware consists of the physical equipment required for system development and evaluation. A detailed discussion of each material, its purpose, and its contribution to the Hivelet system is provided in the succeeding subsections.

3.1.1 Software

The development of the Hivelet system requires modern front-end and back-end technologies to ensure a responsive, accessible, and secure Progressive Web Application (PWA) for both property administrators and tenants of a small-to-medium apartment system.

Table 1. Software Requirements

Software	Version
Visual Studio Code	1.85+
HTML / Tailwind CSS	HTML5 / v3.0+
Express.js	Latest Stable
JavaScript	ES6+ / 0.73+
Node.js	LTS(v18+)
MySQL	8.0

The hardware and software requirements should be in a Windows 10 or higher to ensure long-term system compatibility, stability, and security.

As outlined in Table 1, the development will utilize HTML5, Tailwind CSS, and JavaScript to create a responsive and mobile-adaptive user interface. Node.js and Express.js will serve as the back-end framework responsible for handling server-side logic such as booking management, financial tracking, and issue ticket processing.

MySQL will be used as the system's database management tool to store and organize structured data such as tenant records, transaction history, and system logs. Google Chrome will be the primary browser for development and testing due to its reliability and optimization for modern web applications.

Additionally, Figma will be utilized during the design phase to create user interface prototypes for both administrative and tenant modules prior to actual system development.

3.1.2 Hardware

To ensure efficient development, testing, and deployment of the Hivelet system, specific hardware specifications will be utilized during the development process. Since the system follows a Progressive Web Application (PWA) architecture, both computer workstations and mobile devices will be required.

Table 2. Workstation Hardware Specifications

Component	Minimum Specification
Processor	Intel Core i5 or AMD Ryzen 5
Memory (RAM)	8 GB DDR4
Storage	256 GB Solid State Drive (SSD)
Internet Connection	25 Mbps Broadband/Fiber

As detailed in Table 2, the workstation used for local development, database management, and system testing requires an SSD to ensure fast processing and smooth execution of development tools.

For mobile testing, a mid-range Android device will be used to evaluate system responsiveness and performance under real-world conditions.

As shown in Table 3, this baseline mobile setup ensures the development team can accurately evaluate the system's UI/UX layout and mobile adaptability using modern device standards.

Table 3. Mobile Testing Hardware Specifications (Baseline)]

Component	Version
Operating System	Android 13 or higher
Processor/Chipset	Octa-core (MediaTek Helio G99 or equivalent)
Memory (RAM)	8 GB
Storage	256 GB
Display	6.78-inch FHD+ (For UI/UX layout testing)

3.2 Software Methodology

The researchers will utilize the Agile Software Development Life Cycle (SDLC) in the development of the Hivelet system. Agile is an iterative and flexible software development approach that emphasizes continuous improvement, stakeholder collaboration, and incremental delivery of system components rather than a rigid, linear process (Pressman & Maxim, 2020).

As illustrated in Figure 2, the dynamic and resource-constrained nature of small-scale apartment management requires a development approach that can adapt to changing operational needs. In this context, Agile is selected as the most appropriate methodology as it allows continuous interaction with the landlady and management of the Fe Galang Da Silva Apartment, ensuring that system features are aligned with actual administrative workflows such as tenant management, financial tracking, booking coordination, and maintenance handling.

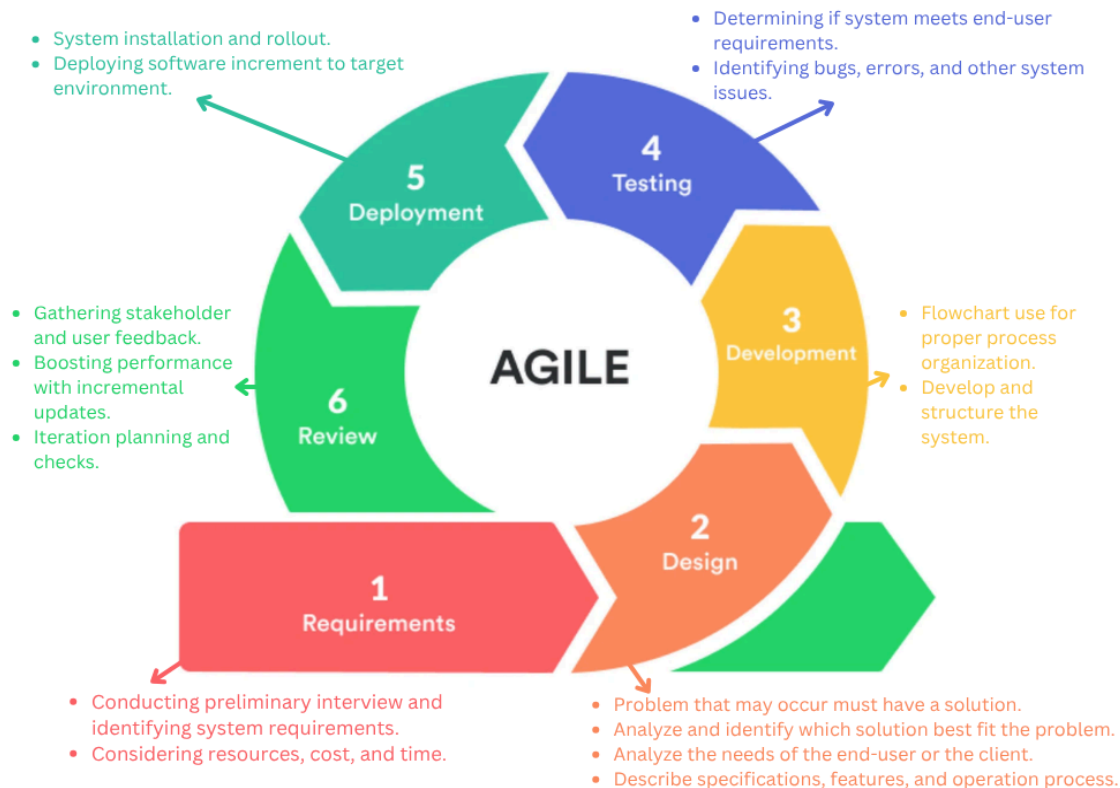


Figure 3. Agile Software Development Life Cycle

Unlike traditional SDLC models, Agile supports early identification of system issues through iterative cycles of development and testing. This enables the researchers to refine system components based on real-time feedback, resulting in a more reliable, user-centered, and context-appropriate solution. Furthermore, the Agile approach facilitates progressive evaluation of the system throughout its development. The system will be assessed using the ISO/IEC 25010 software quality model encompassing functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. This ensures that the developed system not only meets user requirements but also adheres to established software quality standards.

Additionally, the flexibility of Agile allows the system to be continuously improved, modified, and adapted without compromising performance. This is particularly important in ensuring that Hivelet remains functional under varying conditions such as limited infrastructure, unstable internet connectivity, and evolving user needs.

3.2.1 Requirements

In this phase, the researchers conduct needfinding through direct collaboration with the landlady and management of the Fe Galang Da Silva Apartment. This aligns with the first objective of the study, which focuses on identifying user needs and operational challenges. Data is gathered through interviews and observation to examine existing administrative practices, including the use of manual ledgers, fragmented financial tracking, and unstructured communication for maintenance requests. These findings reveal inefficiencies such as inconsistent record-keeping, delayed communication, and lack of centralized monitoring. Based on these insights, the researchers define the system requirements, scope, and core functionalities of Hivelet. Emphasis is placed on the need for a centralized digital platform capable of integrating booking, financial tracking, and maintenance management within a unified system.

3.2.2 Design

During the Design phase, the identified requirements are translated into system architecture and interface prototypes. The researchers develop

wireframes and UI/UX designs using Figma, covering three user roles: public users, tenants, and administrators. The system is designed with consideration for a resource-constrained environment, characterized by limited infrastructure, unstable internet connectivity, and minimal administrative resources. To address these constraints, Hivelet is structured as a lightweight Progressive Web Application (PWA) with optimized performance, simple navigation, and minimal system requirements.

Workflows for booking, financial monitoring, and maintenance requests are mapped and refined through stakeholder feedback to ensure alignment with real-world operations.

3.2.3 Development

This phase corresponds to the second objective of the study, which focuses on developing the system. The researchers implement the Hivelet system using HTML5, Tailwind CSS, and JavaScript for the front-end interface, ensuring responsiveness across devices. The back-end is developed using Node.js and Express.js to manage core functionalities such as booking processing, financial computations, and maintenance ticket handling. An optional online payment feature is implemented as an optional integration to simulate or record digital payments without replacing the primary cash-based financial workflow. MySQL is used as the database management system to store tenant records, transaction data, and system logs. The system is developed

incrementally, following Agile principles, allowing continuous refinement based on feedback and testing results.

3.2.4 Testing

Aligned with the third objective, this phase focuses on testing and evaluating the system's performance. The researchers conduct unit testing to verify individual components such as booking requests, payment tracking, and notification features. Integration testing is also performed to ensure seamless interaction between system modules, including booking, financial management, and maintenance tracking. The optional online payment feature will be tested for basic functionality, including transaction recording and user interaction. Testing ensures that the system meets functional requirements, maintains reliability, and provides a user-friendly experience for both administrators and tenants.

3.2.5 Deployment

After successful testing, the Hivelet system is deployed to a university-managed server environment, where the system will be accessible through a designated URL endpoint for real-world usage and evaluation. A pilot implementation is conducted with selected users to validate real-world performance before full deployment. The researchers perform final system configuration, database migration, and environment setup to ensure smooth operation.

3.2.6 Review and Maintenance

This phase complements the evaluation procedure by implementing refinements and system improvements based on the results gathered from testing and ISO/IEC 25010 evaluation. Following Agile principles, identified issues related to functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability are addressed through iterative optimization and maintenance activities.

3.3 Evaluation Procedure

This phase directly supports the fourth objective of the study, which focuses on evaluating and optimizing the system using the ISO/IEC 25010 Software Quality Model. To ensure a realistic measurement of system performance and usability, the evaluation will be conducted during actual use in the deployment environment, with a specific emphasis on eight key attributes: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability

A structured survey questionnaire is administered to the property owner and tenants of the Fe Galang Da Silva Apartment. Responses are measured using a 5-point Likert scale to assess system effectiveness, user satisfaction, and perceived security.

Statistical analysis is conducted to determine the extent to which Hivelet improves operational consistency, data reliability, and system security in a small-scale apartment setting.

Table 3. Likert Scale

Range	Meaning	Description
5	Strongly Agree	The system has functional completeness of tasks, is effective and reliable.
4	Agree	The system has somehow functional completeness of tasks, is somehow effective and reliable.
3	Neutral	The system has slightly functional completeness of tasks, is slightly effective and reliable.
2	Disagree	The system has slightly no functional completeness of tasks, is slightly not effective and reliable.
1	Strongly Disagree	The system has no functional completeness of tasks, is not effective and reliable.

4 RESULTS AND DISCUSSION

This chapter presents the results and analysis of findings made in the conduct of the study...

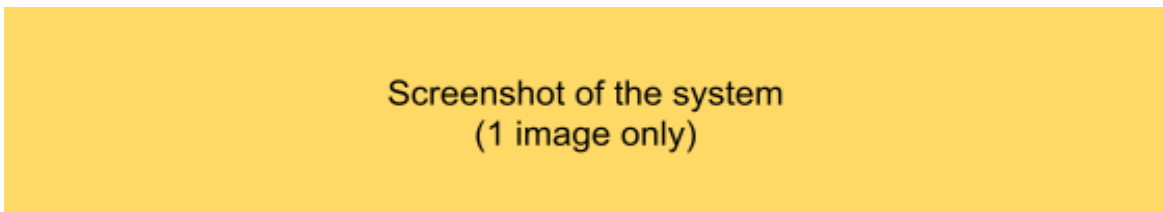
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4.1 Objective 1 Title

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4.1.1 Sub-objective Title

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Figure 2. Sub-objective Title

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4.2 Objective 2 Title

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4.2.1 Sub-objective Title

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4.3 Evaluation of the Developed System

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Evaluation Results

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Deployment Plan and Strategies

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5 SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

This chapter presents the summary, conclusions, and recommendations of the study...

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5.1 Summary

Based on the objectives of the study, the following results were accomplished:

1. The application was developed to provide a platform for auto shops and upgrade the manual process of booking of services. It enables clients to experience the best convenience in repairing and conducting maintenance of cars. Giving a hassle-free experience in locating auto shops near them, booking the services, and providing rescue whenever they have car breakdowns while on the road. The application was designed to be user-friendly, interactive, easy to learn and accessible through a mobile phone that everyone has.

2. The features of the system aid various end-users to accomplish specific tasks. Auto shop owners have been able to facilitate and manage shop personal information, bookings, exchange conversation between customers and view transaction history. The auto shop customer was able to book a service wherever the customer is located as long as the location has cellular signal. Customers were able to exchange conversations with the auto shop owner to check and do follow up on the booked services. The customer's feedback and

rating helped the auto shop to improve and provide the best service for their customer to experience the best quality of service that they are.

3. The system used ISO/IEC 25010 software product quality which was used to evaluate the system's capabilities under the factors of Functional Suitability, Usability, and Performance Efficiency. The evaluation conducted was responded to by the auto shop owners and auto shop customers. The application functionality and features were tested by both auto shop owners and auto shop customers. After the testing phase of the application both auto shop owners and auto shop customers answered a series of questions under the ISO/IEC 25010. The following are the results of the ISO/IEC 25010 characteristics:

- a. The system's functionality usability has an overall rating of 4.98 which means that the application provides ease in using the interface and facilitates the accomplishment of specified tasks and objectives.
- b. Usability got the overall rating of 4.43, which indicates that the application has attributes that make it easy to operate and control. It also shows that the application is appropriate for their needs.
- c. For Performance Efficiency, the overall rating is 4.15 which shows that the application responds and processes the client task immediately. It changes the demands placed by the user in an efficient manner.

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5.2 Conclusions

Based on the analysis done on the results of the study, the following conclusions were drawn:

1. The proponents...

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5.3 Recommendations

Based on the summary and conclusions of the study, the following recommendations...

1. The future researchers...

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APPENDICES

Appendix A: Letter of Request

Appendix B: Appointments

Appendix N: Source Code

CURRICULUM VITAE

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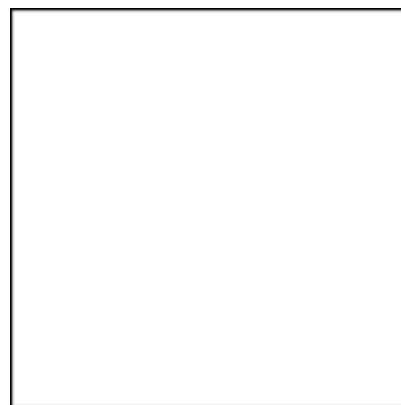
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Sex:

Civil Status:

Email:



Educational Background

Tertiary: Bicol University
 Rizal St. Legazpi City, Albay
 2019-2023

Secondary: Senior High School

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2013-2019

Junior High School

<Address>

2013-2019

Elementary: Elementary School

<Address>

2007-2013

Certificates:

Title
Date
Place

Title
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